Building the Interstellar Agent

Deploying a Single-Task AI Agent with ADK, Gemini, and Cloud Run

The Problem Statement

Build and deploy a single AI agent using ADK and Gemini that is hosted on Cloud Run and performs one clearly defined task. The agent must be callable via an HTTP endpoint and return a valid response for a given input.

The Idea: Interstellar Exploration Bureau

To fulfill the requirements, we designed an educational AI agent specialized in one clear task: acting as a guide to our solar system.

Core Concept:

- Answers specific queries about planets, moons, and missions.
- Uses ADK tools (like Wikipedia integration) to ground responses in real, factual data.
- Provides structured, conversational outputs tailored for space enthusiasts and students.

Meeting the Build Criteria

Single, Defined Task

The agent strictly focuses on planetary education and space facts, refusing off-topic prompts to maintain a tight scope.

Powered by ADK & Gemini

Utilizes the Google Agent Development Kit for state management and tool orchestration, with Gemini providing the conversational intelligence.

Hosted on Cloud Run

Containerized application deployed to Google Cloud Run, ensuring scalable, serverless execution without managing infrastructure.

HTTP Endpoint Integration

Exposes a secure RESTful API. It accepts user input via HTTP POST and returns structured, valid JSON responses.

Future Opportunities

STEM Education

Can be integrated into classroom applications or e-learning platforms to provide interactive, accurate astronomy tutoring for students.

Public Outreach

Museums and planetariums can use the HTTP endpoint to power interactive kiosks, answering visitor questions in real-time.

Reference Architecture

Serves as a robust, open-source template for developers looking to build and scale task-specific ADK agents on Google Cloud.

Features Offered

Dynamic Grounding

By leveraging ADK's built-in tool calling, the agent connects to Wikipedia to fetch up-to-date data, preventing hallucinations and ensuring factual accuracy regarding planetary measurements and events.

Serverless Scalability

Deployed on Cloud Run, the HTTP endpoint can scale from zero to thousands of requests automatically. This ensures high availability and cost-efficiency, only charging for the exact compute time used.

Process Flow Diagram

1. Client Request

User sends a query (e.g., "Tell me about Mars") via an HTTP POST request to the API endpoint.

2. Cloud Run Ingress

Google Cloud Run receives the request, spins up the container, and routes the payload to the ADK application.

3. ADK & Gemini Processing

ADK manages state and invokes the Wikipedia tool. Gemini processes the tool's output to format a coherent answer.

4. Valid Response

The application packages Gemini's output into a structured JSON format and returns it to the client over HTTP.

Technologies Used

Gemini API

Google's advanced LLM provides the natural language understanding and generation capabilities.

Google ADK

The Agent Development Kit structures the logic, handles prompt states, and manages external tool execution.

Cloud Run

A fully managed compute platform that automatically scales the stateless agent container via HTTP requests.

modern chatbot UI dashboard dark mode AI agent

The Agent in Action

The Interstellar Exploration Bureau acts as an interactive bridge between users and complex astronomical data.

When a user prompts the agent for details about a planet, the ADK orchestrates a precise trace. It adds the prompt to the state, transfers control to the agent logic, and invokes external knowledge bases (like Wikipedia) to gather accurate context.

The raw data is then synthesized by Gemini into an easy-to-read format, extracting key metrics like:

- ✓ Size and Gravity
- ✓ Atmosphere and Temperature
- ✓ Surface Features

> The image on the right illustrates the ADK trace interface, showing the successful step-by-step execution and the final formulated response generated by the endpoint.

elopment Kit

Events State Artifacts

SESSION ID ae0c78da-facf-46f4-8d10-9bda16b5951e

Token Streaming | + New

hi

Welcome to the Interstellar Exploration Bureau! I can help you learn about the planets, moons, and missions in our solar system. Where would you like to go today?

mars

add_prompt_to_state

✓ add_prompt_to_state

transfer_to_agent

✓ transfer_to_agent

wikipedia

✓ wikipedia

Mars, often called the "Red Planet" due to its orange-red appearance, is the fourth planet from the Sun. It's a rocky, desert-like world with a thin atmosphere composed mainly of carbon dioxide.

Here are some key facts about Mars:

- **Size and Gravity:** Mars has a diameter of about 6,779 km (4,212 mi), which is roughly half that of Earth. Its surface gravity is about a third of Earth's.
- **Atmosphere and Temperature:** The Martian atmosphere is very thin, with an atmospheric pressure that is only a few thousandths of Earth's. Temperatures on Mars can range drastically from −153 to 20 °C (−243 to 68 °F).
- **Water:** While there are no bodies of liquid surface water, Mars retains some water in the form of ground ice, atmospheric water vapor (forming cirrus clouds, fog, and frost), and large polar ice caps (which also contain seasonal CO_2 snow).
- **Surface Features:** The Martian terrain is divided into a relatively flat, low-lying northern hemisphere and cratered highlands in the southern hemisphere. It features enormous, extinct volcanoes like Olympus Mons (the tallest at 21.9 km or 13.6 mi) and one of the largest canyons in the Solar System, Valles Marineris (4,000 km or 2,500 mi long). Fine dust is prevalent across its surface and in its atmosphere. The planet also experiences "marsquakes."
- **Moons:** Mars has two small, irregularly shaped natural satellites: Phobos and Deimos.

Type a Message...